

Jacen Li

240-589-9362 | jacenli@uw.edu | [linkedin.com/in/jacen-li/](https://www.linkedin.com/in/jacen-li/) | github.com/jtl06 | US Citizen

Education

University of Washington - Seattle

Expected Spring 2028

- BSECE - Embedded Systems - GPA 3.85/4.0
- Courses: Comp Arch I & II, Hardware/Software Interface, Embedded Systems, Digital Design I & II, DSA

Skills

Languages: C++23, C, Python, Assembly (x86-64, ARM, RISC-V), Shell (Bash/Zsh)

Tools: Linux, Git/Github, perf, strace, GCC/Clang, GDB/objdump, Codex/Claude Code, Compiler Explorer

Experience

Arm

Summer 2026

Software Engineering Intern - System Performance, Infra Tooling Solutions

Austin, TX

Developed microbenchmarks for ASCT, the Arm System Characterization Tool

github.com/Arm/asct

- Evaluated 20+ core-to-core latency benchmarks across seven Intel and Arm systems, comparing measurement semantics, stability, sampling strategy, topology coverage, and susceptibility to prefetching.
- Designed directional dependent-pointer-chase and CAS microbenchmarks; used PMU and CMN counters to separate cache-line transfer latency from atomic contention, instruction overhead, and prefetcher effect.
- Mapped multimodal core-to-core latency distributions to physical-address-dependent home-node selection and coherence-path distance, validating relationship with mesh PMU counters and watchpoints.
- Reworked single-cache-line measurements to sample across multiple cache lines, reducing standard deviation from 4.0→0.8 ns on Intel 8275CL and 4.7→0.1 ns on Graviton4 without shifting median latency.
- Developed a homing-aware core-to-core latency benchmark that exposes home-node placement as a third variable and generates heatmaps across source core, target core, and home-node combinations.

National Institute of Standards and Technology

2023

Intern - Sensor Science Division

Gaithersburg, MD

- Deployed RasPi temperature sensor network to profile thermal gradients in the Tape Tunnel test chamber.
- Quantified calibration error propagation w/ Monte Carlo, reduced temperature-induced uncertainty by 38%.

Aeus Keyboards

2020 - 2025

Founder & Operator

jacen.li/keyboards.html

- Founded custom keyboard company with over \$40K revenue; handled design, manufacturing, and logistics.
- Designed PCB in KiCAD and developed C firmware, designed for manufacturing custom CNC milled cases.

Technical Projects

BLT - Better Latency Tester (STM32)

github.com/jtl06/BLT

- Built input-to-photon latency measurement device to benchmark and measure gaming PC responsiveness.
- Utilized timers, circular DMA, and ISRs for us-level resolution, quantified DMA ISR jitter: $\pm 8 \mu\text{s}$ at 2.5 kHz.

Libdivide OSS contribution - Fast Division with ARM SVE

jacen.li/oss/libdivide

- Implemented ARM SVE/SVE2 SIMD paths for libdivide, with tests, documentation, and Github CI fixes.
- Achieved ~5x higher 16bit and ~3x higher 64bit division throughput than NEON on AWS Graviton4.

SwiftFIX - SIMD FIX pre-parser for QuickFIX

github.com/jtl06/SwiftFIX

- Developed C++20 FIX preparer that scans, indexes, validates, and extracts into a zero-alloc offset table.
- Improved frame-splitting throughput from 146 MiB/s to 1.60 GiB/s scalar and 2.19 GiB/s with AVX2

Nycticorax - Discord AI Agent

github.com/jtl06/Nycticorax

- Built a bounded agentic control loop for Discord, w/ semantic memory, context, tools, recovery, and budgets.
- Designed self evaluation benchmarks w/ PostgreSQL telemetry for tool & token use, latency, and failure logs.